

1. Execute Directory-oriented commands: cd, pwd, mkdir, rmdir and their options

Objective:

To equip students with core competencies in navigating and managing the Linux filesystem hierarchy using directory-specific utilities, enabling efficient structure creation, absolute and relative traversal, symbolic link path handling, and safe directory cleanup.

Learning Outcomes

By the end of this lab, students will be able to:

Identify the current working directory path using pwd (in both logical and physical modes).

Navigate the filesystem tree efficiently using standard shortcuts and path arguments with cd.

Construct nested directory structures and set default directory permissions using mkdir.

Remove empty single and nested directory trees safely using rmdir.

Apply command-line options (-p, -m, -v, -L, -P) to control directory operations.

Procedure:

i) pwd (Print Working Directory)

Prints the full absolute path of the current working directory to standard output.

Bash

pwd [OPTION]

- **-L (Logical):** Prints the environment \$PWD value, including symbolic link paths (default behavior).
- **-P (Physical):** Resolves all symbolic links and displays the actual physical directory path on the storage volume.

Example:

Example: If 'link_dir' points to '/var/www/html'

cd link_dir

pwd -L # Output: /home/user/link_dir

pwd -P # Output: /var/www/html

ii. **cd (Change Directory)**

Changes the shell working directory to the target directory specified.

Bash

`cd [OPTION] [DIRECTORY]`

- **cd** or **cd ~**: Changes to the current user's home directory (\$HOME).
- **cd ..**: Moves up one level to the parent directory.
- **cd -**: Toggles back to the previous working directory recorded in \$OLDPWD.
- **cd /**: Switches directly to the root directory.
- **-P**: Forces cd to navigate through physical directory paths rather than following symbolic links.

iii. **mkdir (Make Directories)**

Creates one or more new directories at the specified paths.

Bash

`mkdir [OPTIONS] DIRECTORY...`

- **-p (--parents)**: Creates parent directories as needed without throwing an error if they already exist. Essential for building deep paths in one command.

```
mkdir college
mkdir college/linux
mkdir college/linux/commands
mkdir college/linux/commands/week1

mkdir -p college/linux/commands/week1

ls -R college
```

- **-m (--mode)**: Sets directory permission bits (like chmod) at creation time rather than applying the system umask.
- **-v (--verbose)**: Displays a message for every created directory.

```
mkdir -v first second three
```

Bash

Creates deep subfolder structure and sets permissions to 755 (rwxr-xr-x)

```
mkdir -pm 755 app/build/logs/2026
```

iv. **rmdir (Remove Empty Directories)**

Deletes empty directories. It acts as a safety tool because it refuses to delete folders containing files or other subdirectories.

Bash

```
rmdir [OPTIONS] DIRECTORY...
```

- **-p (--parents):** Removes the specified directory and its empty parent directories recursively.
- **-v (--verbose):** Outputs diagnostic information for every directory processed.
- **--ignore-fail-on-non-empty:** Suppresses failure messages if a directory cannot be removed because it contains files.

Safely removes 'nested', then 'parent', then 'grandparent' if all are empty

```
rmdir -pv grandparent/parent/nested
```